CHEE 4001

DP 2 FOR WASTE TREATMENT

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I-core 2.1

I-core 2.1.1 Equipment selection, design and sizing

I-core 2.1.1.1 Absorption column T-501

I. Design Inputs (T-501)

Process objective	Removal of NOx from gaseous fluids		
Basis for design	NOx gas needs to be removed to meet the requirements of the Queensland Environment Act.		
Scientific principle	Aspen / For tray sizing, calculations are performed using the methods provided in Perry's Handbook and by Kister and Haas. For tower height calculations, the method used is from James R. Couper (2010).		
Major input streams	Gas		
	Fluid descriptions	Waste gas	
	Temperature(° C)	130.49	
	Pressure(bar)	1.1	
	Flow rate(cum/hr)	615052	
	Mass flow rate(kg/hr)	56424.52	
	Concentration	H2O: 0.006 N2: 0.912 O2: 0.081 NO: 3*10^(-11) NO2: 3.26*10^(-28) N2O: 3.15*10^(-6)	
	Viscosity(cp)		
	Liquid(contains recycle and H2SO4)		
	Fluid descriptions	Waste liquid(contains circular flow)	Liquid used for absorption
	Temperature(° C)	Recycle: 42	H2SO4: 25.16
	Pressure(bar)	2	1
	Flow rate(cum/hr)	56.771	0.921
	Mass flow rate(kg/hr)	57980.5	1280.21
	Concentration(major)	H2O: 0.879 NH4+: 0.017 NO: 2.59*10^(-13) NO2: 0 N2O: 5.943*10^(-9)	H2O: 0.408 HSO4-: 0.495 H3O+: 0.097
	Viscosity(cp)		
Major equations	In tray sizing: Using flow rate and surface tension, using Kister and Haas equation, C-Factor method In tower height: Using the concentration and flow rate , using karren' 3004 lec gas-liquid		

II. Equipment specification sheets for Out design (T-501)

Outlet streams	
Gas outlet flow rate	58773.99 (kg/hr)
Gas outlet composition(major)	H2O: 0.046 N2: 0.876 O2: 0.078 N2O: 2.207*10^(-11)
Liquid outlet flow rate	56912.61 (kg/hr)
Liquid outlet composition	H2O: 0.864 N2O: 6.808*10^(-11) SO4--: 0.0224 HSO4-: 0.0889
Assumptions	Basic on the aspen
Height of column	3.14 (meter)
Column diameter	2.721 (meter)
Overall pressure drop	0.0548 (bar)
Equipment Parameter Specifications	
Tray layouts in columns	Single-pass crossflow trays
Spacing time	0.0668 (hr)
Decks	Sieve (cost low)
Materials of construction	316 stainless steel
Number of stages	5
Weirs	Notched weir
Tray spacing	0.6096 (meter)
Flooding rate	76.0009 (%)
liquid phase volume rate/tower area	7.4 (cum/hr/sqm)
Total head loss (thermal liquid height)	0.515 (meter)
Wall thickness	0.003 (meter)

III. Calculations and justification (T-501)

Two methods were used to calculate the diameter and height of the adsorption tower: Aspen and manual calculation (mainly using the C-method and Kister equation). In addition, since manual calculations require assumptions, assumptions were also made regarding orifice size and tray spacing, as follows:

Assumption	Description
1	Since there are multiple feed streams, the one with the largest flow rate was chosen as the basis for the design. For example, the surface tension coefficient used is that of fluid 515, while the orifice size and tray spacing are taken from the data in Aspen.
2	No heat loss
3	No change in physical properties. (eg.density)
4	derating factor = 0.85 (Use this value for initial calculations) flood factor = 1 (Use this value for initial calculations)

Material Selection: 316 stainless steel was chosen as the material because, according to the corrosion chart from the British Stainless Steel Association, 304 stainless steel is prone to pitting corrosion under medium concentration conditions. Therefore, 316 stainless steel was selected for its superior resistance to corrosion.

Tray layouts: A common tray design is used, which provides high capacity and efficient vapor-liquid contact area.

Summarize: After manual calculation, the final column diameter was determined to be 2.669 meters, which is close to the value obtained from Aspen. Considering that the manual calculation used simplified methods, such as not accounting for pressure drop and side weir loading, which can affect the design, the result from Aspen was chosen.(The calculation steps are in Appendix 1) (All calculation from James R. Couper ,.2010)

I-core 2.1.1.2 Heat exchanger for E-501

I. Design Inputs (E-501)

Process objective	Cooling circulating stream, consistent with absorbent temperature
Basis for design	Cooling circulation flow from 50 to 42°C
Scientific principle	Aspen / calculation basic on Towler (2022)

Stream input	Liquid		
	Fluid descriptions	Hot liquid	Cold liquid
	Temperature(° C)	50.05	25
	Pressure(bar)	2	1
	Flow rate(cum/hr)	57.09	41.71
	Mass flow rate(kg/hr)	57980.45	41508.71
	Specific heat capacity kJ/(kg·K)	3.96	4.189
	Latent Heat of Vaporization(kcal/kg)	No	-52765.06
Major equations	For area: Q = U*A*delt Tlm		

II. Equipment specification sheets for Out design (E-501)

Heat exchanger type	Shell and tube	
TEMA type	BEM	
	Shell side	**Tube side**
Fluid placement	Water	Waste water(circulation flow)
Number passes	1	2
Overall heat transfer coefficient U (W/m^2*C)	1150	
Fouling factor(W/m^2*C)	7000	5000
Transfer area (m^2)	38	
Materials of construction	Stainless steel (316 vertical)	Stainless steel (316 vertical)

III. Calculations and justification (E-501)

The shell and tube heat exchanger was chosen because the required surface area is not large, and the fluids involved are common, so no special design is needed. A 1-shell side and 2-tube side configuration was selected as a simple design. (same with E-502)

Material Selection: 316 stainless steel was chosen because the fluid contains some sulfuric acid, and 316 stainless steel is used to prevent corrosion.

Calculation steps and detailed options are in Appendix 2

I-core 2.1.1.3 Heat exchanger for E-502

I. Design Inputs (E-502)

Process objective	As a condenser, it cools the exhaust gas to a liquid, which then enters a subsequent collection tank		
Basis for design	Cools down to 42°C while condensing gas		
Scientific principle	Aspen / calculation basic on Towler (2022)		
Stream input	Liquid		
	Fluid descriptions	Hot liquid(waste gas from section 3 Evaporator)	Cold liquid(cooler water)
	Temperature(° C)	83.08	25
	Pressure(bar)	0.2	1
	Flow rate(cum/hr)	56.771	444.04
	Mass flow rate(kg/hr)	7511.56	441816
	Specific heat capacity kJ/(kg·K)	3.87	4.189
	Latent Heat of Vaporization(kcal/kg)	No	-52765.06
Major equations	For area: Q = U*A*delt Tlm		

II. Equipment specification sheets for Out design (E-502)

Heat exchanger type	Shell and tube	
TEMA type	BEM	
	Shell side	**Tube side**
Fluid placement	Water	Waste gas
Number passes	1	2
Overall heat transfer coefficient U (W/m^2*C)	1150	
Fouling factor(W/m^2*C)	7000	5000
Transfer area (m^2)	184	
Materials of construction	Steel	Steel

III. Calculations and justification (E-502)

Material Selection: No special material selection is required, so the use of steel

The calculation method and design will follow the steps outlined in Towler (2021). The main equation used is Q = U*A*delt Tlm, where Q is taken from the data provided by Aspen. The relevant specific heat capacities and other coefficients are also sourced from Aspen. (For detailed steps, please refer to Appendix 2)

I-core 2.1.1.4 Pump for P-501

I. Equipment specification sheets for design (P-501)

Input design	
Flow rate(cum/hr)	57.09
Temperature(° C)	50.04
Pressure(bar)	2
Fluid descriptions	Waste water(circulation flow)
Mass flow rate(kg/hr)	57980.45
Density(kg/m^3)	1015.53
Viscosity(cp)	0.61
Average molecular mass,	19.59
Pump efficiency assumed	0.64 (basic on aspen)
Output design	
Pump type	Centrifugal
Duty (KW)	2.46
Materials of construction	Stainless steel (316 vertical)
Safety and hazard	Need corrosive resistant rotor

II. Calculations and justifications (P-501)

Pump type: The fluid here is a circulating flow, so a pump is used for pressurization. A centrifugal pump was chosen due to its simple design, wide applicability, and suitability for low-viscosity liquids, allowing for efficient operation.

Material Selection: Since the fluid contains sulfuric acid, 316 stainless steel was chosen to prevent corrosion.

I-core 2.1.1.5 Compressors for P-502

I. Equipment specification sheets for design (P-502)

Input design	
Flow rate(cum/hr)	8.16
Temperature(° C)	42.14
Pressure(bar)	1.2
Fluid descriptions	Waste gas from section 3 Evaporator
Mass flow rate(kg/hr)	7511.57
Density(kg/m^3)	920.28
Viscosity(cp)	0.016
Average molecular mass,	18.01
Output design	
Compressor type	Centrifugal
Duty (KW)	0.89
Materials of construction	Steel
Compressor efficiency	0.48 (basic on aspen)
Safety and hazard	No

II. Calculations and justifications (P-502)

Pump type: A centrifugal was chosen due to its simple design, wide applicability, and suitability for low-viscosity liquids, allowing for efficient operation.

Material Selection: Only ordinary steel is required.

I-core 2.1.1.6 Turbine for TB-501

I. Equipment specification sheets for design (TB-501)

Input design	
Flow rate(cum/hr)	61051.96
Temperature(° C)	130.49
Pressure(bar)	1.1
Fluid descriptions	Waste gas from section 1 HNO3 Reactor

Mass flow rate(kg/hr)	56424.52
Density(kg/m^3)	0.92
Viscosity(cp)	0.022
Average molecular mass,	28.20
Output design	
Compressor type	Turbines
Duty (KW) (from aspen)	3467.56
Materials of construction	Steel
Safety and hazard	No
Efficiency (%)	85
Generated electrical energy (Gj/yr)	92810

II. Calculations and justifications (TB-501)

No design is required here, based on the selection and valuation of relevant equipment on the market

I-core 2.1.1.7 Storage Tank for T-502

I. Equipment specification sheets for design (T-502)

Input design	
Time required for storage	Two weeks
Storage volume required (total)	430150.56 (kg)
Temperature	25 (C)
Pressure	1 (bar)
Fluid description	50% sulfuric acid solution
Density	1389.8 (kg/cum)
Viscosity	4.004 (cp)
Phase of inlet	Liquid
Phase of outlet	Liquid
Output design	
Tank type	Vertical cylindrical with fixed-roof

Design pressure	1.1 (bar)
Materials of construction	PVC with rubber lining
Process internals	Agitator
Volume of tank	356 (cum)
Number tanks required	1
Diameter of tank	6.1 (m)
Height of tank	12.2 (m)
Maximum liquid level in the tank (freeboard)	10%
Additional requirement	No

II. Calculations and justification (T-502)

Tank type: According to public reports from Poly Processing companies, this tank type is commonly used. In addition, since the stored fluid is a sulfuric acid solution, it tends to absorb moisture from the air and is corrosive. To reduce risks, a fixed-roof design is used

Material Selection: According to Marshall 2022, there are usually three concentration ranges for sulfuric acid storage, and in this case the concentration is 50%, which does not require the use of specific steel, so PVC storage tanks were chosen.

Process internals: Only need agitator, and no heating coil is required; storage at ambient temperature is sufficient. Tank volume calculations and diameter and height selections are in Appendix 3. The calculation method and design will follow the steps outlined in Towler (2021)

I-core 2.1.1.8 Storage Tank for V-501

I. Equipment specification sheets for design (V-501)

Input design	
Time required for storage	1 day
Storage volume required (total)	1546178.632 (kg)
Temperature	50 (C)
Pressure	1 (bar)
Fluid description	Absorbent solution (sulfuric acid solution) and wastewater from section 3
Density	1015.53 (kg/cum)
Viscosity	0.63 (cp)
Phase of inlet	Liquid
Phase of outlet	Liquid
Output design	
Tank type	Vertical cylindrical with fixed-roof
Design pressure	1.1 (bar)
Materials of construction	PVC with rubber lining
Process internals	Agitator and cooler coils
Volume of tank	2018 (cum)
Number tanks required	1
Diameter of tank	12.2 (meter)
Height of tank	15.2 (meter)
Maximum liquid level in the tank (freeboard)	10%
Additional requirement	No

II. Calculations and justifications (V-501)

Time required for storage: Considering that the purpose of the collection tank here is in mixing the recirculation stream and the waste liquid from section 3, it is not necessary to maintain a two-week storage capacity as in T-502, but only a 1-day storage is needed.

Tank type: A fixed-roof design is used because the waste liquid from Section 3 may contain some waste gases, making a sealed storage tank necessary.

Material Selection: On one hand, PVC storage tanks are relatively inexpensive. On the other hand, since the liquid contains sulfuric acid, which is corrosive, PVC material is used for its resistance to corrosion.

Process internals: Stirrers and cooling devices are needed, sulfuric acid mixed with liquids is exothermic, and to prevent the temperature from getting too high, cooling devices are designed to maintain the temperature

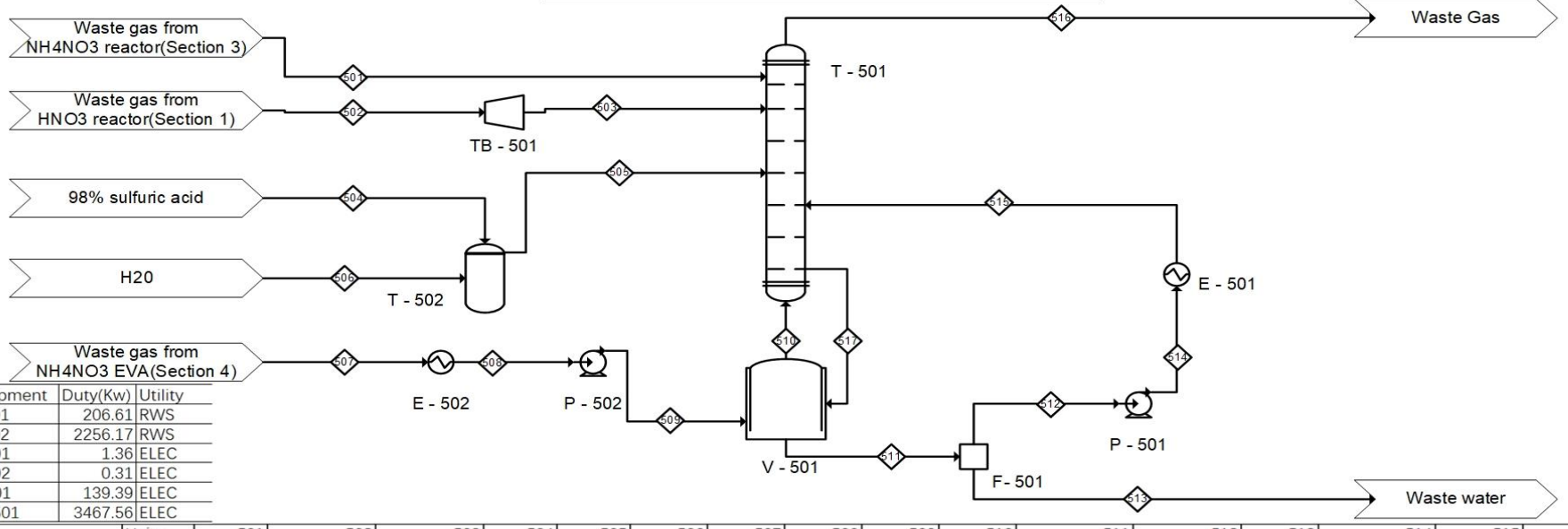
Tank volume calculations and diameter and height selections are in Appendix 3. The calculation method and design will follow the steps outlined in Towler (2021)

I-core 2.1.2 PFD

CP-501 has been changed to P-502 because the initial design confused the physical state of the stream, mistakenly assuming it was gas. Additionally, Section 4 has been significantly updated, with more changes in the exhaust gases. However, the exhaust contains only trace amounts of NH3, which comply with Queensland's emission standards. Therefore, the decision was made not to introduce the updated dryer air into Section 5. For more details, please refer to Section 4. Furthermore, including the updated exhaust would significantly increase costs, making it an uneconomical and unnecessary option.

EQUIPMENT LIST

TB-501	Steam turbine
T-501	Sulfuric acid adsorption tower
T-502	Sulfuric Acid Dilution Tower
E-501	Reactor Effluent Cooler
E-502	Reactor Effluent Cooler
P-501	Pressurized pump
P-502	Pressurized pump
V-501	Wastewater storage tanks
F-501	Liquid separator



Equipment	Duty(Kw)	Utility
E-501	206.61	RWS
E-502	2256.17	RWS
P-501	1.36	ELEC
P-502	0.31	ELEC
V-501	139.39	ELEC
TB-501	3467.56	ELEC

	Units	501	502	503	504	505	506	507	508	509	510	511	512	513	514	515	516	517
Temperature	C	120.00	339.62	130.49	25.00	25.16	25.00	83.08	42.00	42.08		50.00	50.00	50.00	50.05	42.00	62.61	55.60
Pressure	bar	1.00	11.00	1.10	1.00	1.00	1.00	0.20	0.20	1.20	1.00	1.00	1.00	1.00	2.00	2.00	0.90	0.92
Mass Vapor Fraction		1.00	1.00	1.00	0.00	0.00	0.00	1.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	1.00	0.00
Mass-liquid fraction		0.00	0.00	0.00	1.00	1.00	1.00	0.00	1.00	1.00		1.00	1.00	1.00	1.00	1.00	0.00	1.00
Mass solid fraction		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00
Mass flow rate	kg/hr	4236.23	56424.52	56424.52	653.16	1280.21	627.05	3278.30	3278.30	3278.30		31503.40	28353.06	3150.34	28353.06	28353.06	62069.07	28225.10
Component flow	kg/hr																	
H2O	kg/hr	4152.45	360.09	360.09	13.06	640.10	627.05	3254.76	3254.54	3254.54		23768.56	21391.70	2376.86	21391.81	21375.17	6004.12	20519.03
NH3	kg/hr	82.98	0.00	0.00	0.00	0.00	0.00	23.54	23.33	23.33		0.00	0.00	0.00	0.00	0.00	0.00	0.00
HNO3	kg/hr	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00
N2	kg/hr	0.52	51488.66	51488.66	0.00	0.00	0.00	0.00	0.00	0.00		2.40	2.16	0.24	2.16	2.16	51488.94	2.40
O2	kg/hr	0.28	4575.63	4575.63	0.00	0.00	0.00	0.00	0.00	0.00		0.39	0.35	0.04	0.35	0.35	4575.87	0.39
NO	kg/hr	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00
NO2	kg/hr	0.00	2.67	2.67	0.00	0.00	0.00	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00
N2O	kg/hr	0.00	0.13	0.13	0.00	0.00	0.00	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.01	0.00
H2	kg/hr	0.00	0.02	0.02	0.00	0.00	0.00	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.02	0.00
H2SO4	kg/hr	0.00	0.00	0.00	640.10	640.10	0.00	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00

REV	DATE	DESCRIPTION	BY	CHECKED	DATE	APPROVED	DATE
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Create change

Waste gas treatment design - PFD

Status ISSUED FOR REVIEW | DRAWING No PFD | Date 27/08/24 | Rev A

I-core 2.2

I-core 2.2.1 Equipment list

ID	Description	Capital cost (AUD)
T-501	Absorption tower	4,741,190
T-502	Sulfuric acid storage tanks	1,194,196
E-501	Cooler	832,574
E-502	Cooler	393,705
P-501	Booster pump	181,354
P-502	Compressor	14,776
V-501	Circulating flow collection tanks	3,543,459
TB-501	Turbine	2,205,514

I-core 2.2.2 Design table

ID	Cost basis	Bare module cost (AUD)	Material factor
T-501	Height of column =3.14 meter Column diameter = 2.721 meter Wall thickness = 0.003 (meter) Tray number = 5	802,126	1.3
T-502	Flow rates =0.00026 (cum/sec) Freeboard =10 (%) Storage days = 14	224,485	0.9
E-501	Heat transfer area = 38 (cum)	108,351	1.3
E-502	Heat transfer area = 184 (cum)	66,608	1
P-501	Flow rates = 15.86 (L/sec)	23,601	1.3
P-502	= 0.89	2,500	1
V-501	Flow rates = 0.0176 (cum/sec)	666.102	0.9
TB-501	Power duty = 3467.56 (KW)	373,134	1

I-core 2.2.3 Calculation steps and relative costs

Long factor: 4.74, The price Long factor obtained from the aspen plus simulation is 5, so according to the textbook, the factor used is 4.74.

Location factor: The 2003 location factor for Australia was 1.21.The exchange rate in 2003 averaged about 1 AUD =0.65USD, and in 2024 it averaged about 1 AUD=0.67 USD. The 2024 location factor for Australia's thus 1.21 X0.67/0.65=1.247.

Tower: According to Towler (2022), the tower diameter is first calculated by determining the wall thickness and shell mass, which allows for an estimate of its price. Since there are 5 trays, the cost of the trays is also calculated. The total price is obtained by adding the costs of the shell and the trays.

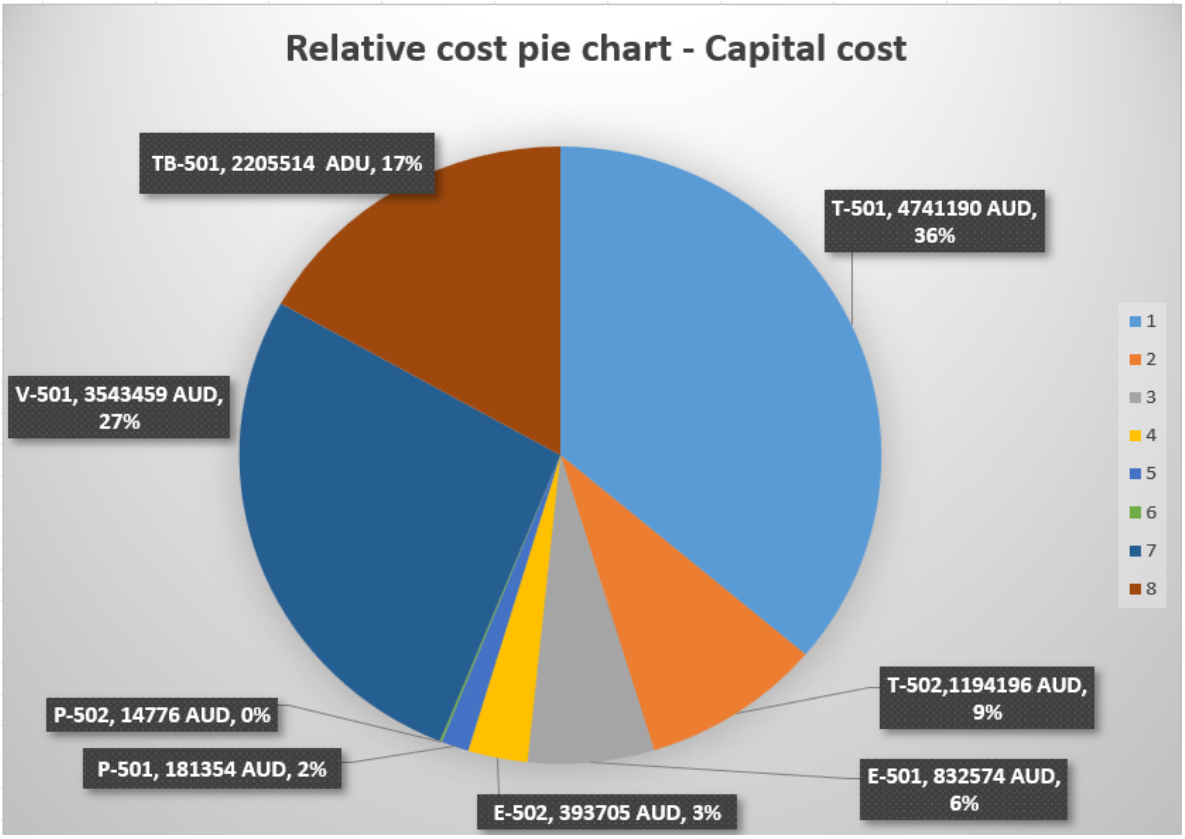
Heat exchanger: Using the estimated price of the transfer area calculated in I.core 2.1, calculate the price according to the values given in table 7.2 in Towler.2022.

Pump: Estimated prices using flow rates through pump, calculating prices according to the values given in table 7.2 in Towler.2022. And for pump 502, its duty is too small, choose to buy from the heat exchanger contractor, choose the cheaper pump, price = 2500 AUD

Tank: Using the volume of the tank calculated in I.core 2.1, calculate the price according to the values given in table 7.2 of Towler .2022 in sequence.

Turbine: Using the value of duty given by Aspen, calculate the price in order of the values given in table 7.2 of Towler .2022. However, after detailed calculations, the result significantly exceeded the price provided by Aspen. As a result, the price was inquired from contractors, and a lower-priced 4000 kW turbine from China's Xinning Energy was selected. This turbine has an efficiency of 85% and is priced at approximately 250,000 USD, which is equivalent to 373,134 AUD. Therefore, it was chosen as the equipment price.

The final relative cost pie chart is calculated by the above steps: (All calculations are sourced from Towler,2021)



I-core 2.3

I-core 2.3.1 Control loop table

I-core 2.3.1.1 E-502

Equipment number	Heat exchanger E-502		
Control loop functional description	Feedback temperature control for heat exchanger, and the waste gases from section 3 need to be condensed while mixing with the subsequent recirculation stream to maintain a fixed temperature.		
Specific control objectives	Environmental protection,		
Control variable (CV)	Outlet process fluid temperature		
Measured variable (MV)	Outlet process fluid temperature		
Manipulated variable (MPV)	The flowrate of cooling water		
Possible disturbance variables	● Changes in inlet process fluid temp and flow (507) ● Water temperature, equipment fouling/ failur		
Control heuristic	CV	MV	MPV
	● CV1: E502 outlet temperature is not self-regulating, requires external control	● MV1. Controlled variable (feedback) ● MV2: Easy to measure (by outstream)	● MPV1. Direct action: The cooling water acts directly on the process stream ● MPV4: Minimize interactions: Move the disturbance downstream

I-core 2.3.1.2 P-502

Equipment number	Booster pump P-502		
Control loop functional description	By using a bypass loop to regulate the fluid pressure entering V-501		
Specific control objectives	Smooth operation		
Control variable (CV)	Outlet process fluid pressure		
Measured variable (MV)	Outlet process fluid pressure		
Manipulated variable (MPV)	The opening of the bypass valve, which controls the flow rate and pressure by adjusting the bypass valve.		
Possible disturbance variables	● Changes in inlet process fluid temp and flow (508) ● Bypass valve lag or failure		
Control heuristic	CV	MV	MPV
	● CV1: The pumping system is not self-regulating and cannot automatically maintain stable pressure	● MV1. The controlled variable is the pressure at the pump outlet (Bypass)	● MPV1. The opening of the bypass valve can directly affect the fluid. ● MPV2: Adjustment should be sensitive and quick to respond

		• MV2: Easy to measure (by outstream) • MV3: The system is able to respond quickly to changes	• MPV3: Hysteresis can lead to pressure control failure

I-core 2.3.1.3 TB-501

Equipment number	Turbine TB-501		
Control loop functional description	To ensure the proper operation of the turbine(TB-501), feedforward control is used to regulate the flow rate entering the turbine(TB-501)		
Specific control objectives	Safety		
Control variable (CV)	Inlet flowrate of process fluid		
Measured variable (MV)	Inlet flowrate of process fluid		
Manipulated variable (MPV)	The opening of the valve		
Possible disturbance variables	• Changes in inlet process fluid temp and flow (502) • Valve lag or failure		
Control heuristic	CV	MV	MPV
	• CV1: The flow rate does not remain at a constant value by itself. • CV2: The flow rate into the turbine must be within certain limits to ensure the safety and efficiency of the equipment	• MV1. The controlled variable is the flowrate of the inlet stream(feedforwards) • MV2: Easy to measure (by inlet stream)	• MPV1. The opening of the valve can directly affect the fluid. • MPV2: Adjustment should be sensitive and quick to respond • MPV3: Hysteresis can lead to pressure control failure

I-core 2.3.1.4 T-502

The tank contains three control loops, two for feed flow control and one for level control, which will be explained in two tables. And the design concept comes from Moe Toghraei Toghraei, 2019

Equipment number	**T-502 for flowrate**	
Control loop number	Feedback Loop 504	Feedback Loop 506
Control loop functional description	To ensure a steady flow of sulfuric acid (504) into the tanks	To ensure a steady flow of H2O (506) into the tanks
Specific control objectives	Product Quality	Same
Control variable (CV)	Liquid flow rate into the tank	Same

Measured variable (MV)	Liquid flow rate into the tank	Same
Manipulated variable (MPV)	The opening of the valve	Same
Possible disturbance variables	● Valve lag or failure ● Change in concentration of sulfuric acid feedstock	● Valve lag or failure ● Cooling water flow is reduced because other units are in use
CV (heuristic)	● CV1: The flow rate does not remain at a constant value by itself.	Same
MV (heuristic)	● MV1: The controlled variable is the flowrate of the outlet stream. ● MV2: Easy to measure (by outlet stream) ● MV3: The flow sensor must be sensitive, because here it is necessary to ensure that the ratio of sulfuric acid to water entering the tank is the same	Same
MPV (heuristic)	● MPV1: The opening of the valve can directly affect the fluid. ● MPV2: Adjustment should be sensitive and quick to respond ● MPV3: Hysteresis can lead to pressure control failure	Same

Equipment number	**T-502 for liquid level**		
Control loop functional description	Feedback control using high and low level sensors to ensure safe tank levels		
Specific control objectives	Safety		
Control variable (CV)	The level of a tank		
Measured variable (MV)	The level of a tank		
Manipulated variable (MPV)	The opening of the valve		
Possible disturbance variables	● Changes in inlet process fluid flow (504 and 506) ● Valve lag or failure ● Leaks from storage tanks		
Control heuristic	CV	MV	MPV
	● CV1: Not self-regulatory	● MV1: The controlled variable is the flowrate of the outlet stream ● MV2: Easy to measure (by outlet stream)	● MPV1: The opening of the valve can directly affect the fluid. ● MPV4: The flow rate at the outlet affects the efficiency of the subsequent absorption tower.

I-core 2.3.1.5 V-501

As a collection tank, the V-501 serves to buffer and mix the absorbent and waste liquids, so it was chosen not to have a flow rate control for the feed stream, but only to have a level control system. And the design concept comes from Moe Toghraei Toghraei, 2019

Equipment number	**V-501 for liquid level**		
Control loop functional description	Feedback control using high and low level sensors to ensure safe tank level.		
Specific control objectives	Safety		
Control variable (CV)	The level of a tank		
Measured variable (MV)	The level of a tank		
Manipulated variable (MPV)	The opening of the valve		
Possible disturbance variables	● Changes in inlet process fluid flow (509 and 517) ● Valve lag or failure ● Leaks from storage tanks		
Control heuristic	CV	MV	MPV
	● CV1: Not self-regulatory	● MV1: The controlled variable is the flowrate of the outlet stream ● MV2: Easy to measure (by outlet stream)	● MPV1: The opening of the valve can directly affect the fluid. ● MPV4: The flow rate at the outlet affects the efficiency of the subsequent absorption tower.

I-core 2.3.1.6 F-501

The splitter is set up to prevent excessive flow in the process, and therefore, two control loops are established to control the flow ratio. And the design concept comes from Moe Toghraei Toghraei, 2019

Equipment number	**F-501 for ratio**	
Control loop number	Feedforward Loop 512	Feedforward Loop 513
Control loop functional description	Use feed-forward to control the flow ratio of recirculating stream and waste liquid to 9:1	Same
Specific control objectives	Safety	Same
Control variable (CV)	Liquid flow rate of 512	Liquid flow rate of 513
Measured variable (MV)	Liquid flow rate of 512	Liquid flow rate of 513
Manipulated variable (MPV)	The opening of the valve	Same
Possible disturbance variables	● Valve lag or failure	● Valve lag or failure
CV (heuristic)	● CV1: The flow rate does not remain at a constant value by itself.	Same
MV (heuristic)	● MV1: The controlled variable is the flowrate of the outlet stream.	Same

	● MV2: Easy to measure (by outlet stream) ● MV3: The flow sensor must be sensitive, because here it is necessary to ensure that the ratio of recirculating stream to waste water is 9:1	
MPV (heuristic)	● MPV1: The opening of the valve can directly affect the fluid. ● MPV2: Adjustment should be sensitive and quick to respond	Same

I-core 2.3.1.7 E-501 and P-501

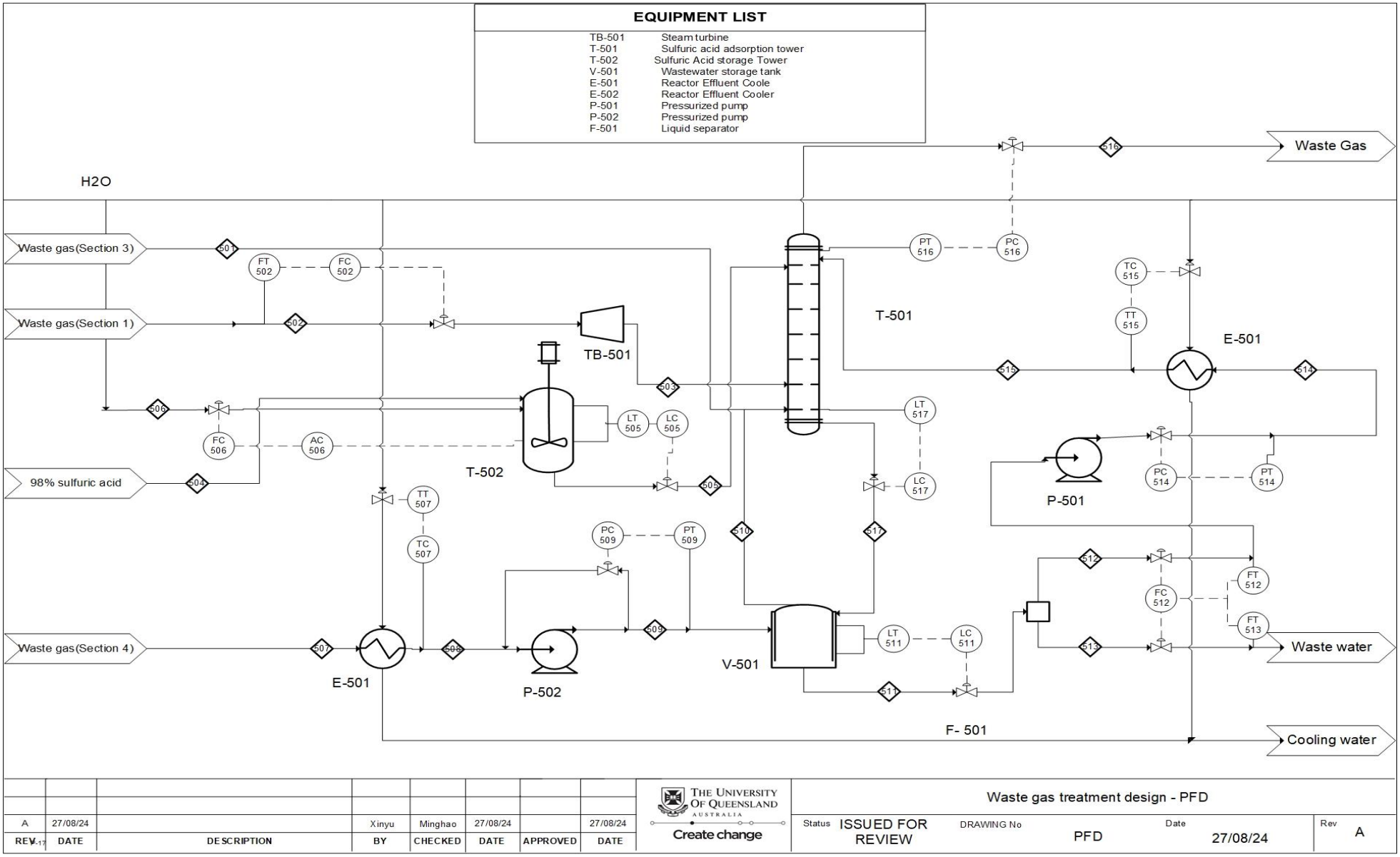
Equipment number	P-501	E-501
Control loop number	Bypass Loop 514	Feedback Loop 515
Control loop functional description	Controls the pressure of the circulating stream into the absorber tower, which needs to be pressurized because of the pressure drop caused by the absorber tower	Adjust temperature to set temperature 42
Specific control objectives	Product Quality	Product Quality
Control variable (CV)	Outlet process fluid pressure (514)	Outlet process fluid temperature (515)
Measured variable (MV)	Outlet process fluid pressure (514)	Outlet process fluid temperature (515)
Manipulated variable (MPV)	The opening of the bypass valve, which controls the flow rate and pressure by adjusting the bypass valve.	The opening of the valve
Possible disturbance variables	● Changes in inlet process fluid temp and flow (512) ● Bypass valve lag or failure	● Valve lag or failure ● Cooling water flow is reduced because other units are in use
CV (heuristic)	● CV1: The pumping system is not self-regulating and cannot automatically maintain stable pressure	● CV1: The flow rate does not remain at a constant value by itself.
MV (heuristic)	● MV1. The controlled variable is the pressure at the pump outlet (Bypass) ● MV2: Easy to measure (by outstream) ● MV3: The system is able to respond quickly to changes	● MV1. Controlled variable (feedback) ● MV2: Easy to measure (by outstream)
MPV (heuristic)	● MPV1. The opening of the bypass valve can directly affect the fluid. ● MPV2: Adjustment should be sensitive and quick to respond ● MPV3: Hysteresis can lead to pressure control failure	● MPV1. Direct action: The cooling water acts directly on the process stream ● MPV4: Minimize interactions: Move the disturbance downstream

I-core 2.3.1.8 T-501

Equipment number	T-501	
Control loop number	Feedback Loop 516	Feedback Loop 517
Control loop functional description	Control of tower pressure to prevent over pressurization in the absorber tower	The liquid level control in the absorption tower requires the bottom

		liquid level to be stabilized to prevent it from rising too high.
Specific control objectives	Safety	Equipment protection
Control variable (CV)	Outlet process gas pressure (516)	liquid level at the bottom of the tower (517)
Measured variable (MV)	Outlet process gas pressure (516)	liquid level at the bottom of the tower (517)
Manipulated variable (MPV)	Gas flow rate controlled by the overhead gas valve	Liquid flow rate controlled by the bottom product valve
Possible disturbance variables	● A partial or complete blockage in the vent or gas outlet line would restrict gas flow, causing a pressure build-up inside the column. (516)	● Foaming inside the column can cause artificial elevation of the liquid level, leading to inaccurate level readings. Fouling can cause flow restrictions that may impact liquid drainage and raise the liquid level.
CV (heuristic)	● CV1: The pressure in the tower is non-self-regulatory because, without active control, the pressure would fluctuate and affect the process.	● CV1: The liquid level in the tower is non-self-regulatory ● CV3: The liquid level affects the gas-liquid interface and therefore the overall mass transfer efficiency in the absorption column
MV (heuristic)	● MV1. The pressure is the variable being measured and controlled (feedback) ● MV2: Easy to measure (by out gas) ● MV3: The pressure sensor is located at the top of the tower to ensure minimal delay in detecting pressure changes	● MV1. Controlled variable (feedback) ● MV2: Easy to measure (by tower liquid level)
MPV (heuristic)	● MPV1. The manipulated variable is the flow rate of overhead gas.	● MPV1. Direct action: the manipulated variable is the flow rate of the bottom product

I-core 2.3.2 PFD



I-core 2.4

I-core 2.4.1 Introduction

The waste gas treatment section of the plant is designed to efficiently remove nitrogen oxides (NOx and N_2O) from the waste gases generated during ammonium nitrate production using a 50% sulfuric acid solution. This system successfully meets the stringent environmental standards set by Queensland's Environmental Protection regulations, limiting NOx emissions to within 0.12 ppm and N_2O emissions to below 9 ppm. Key equipment, such as the absorption tower (T-501), heat exchangers (E-501, E-502), and pumps, were selected and designed to meet these objectives.

The absorption tower, the most critical component of the system, is designed to maximize gas-liquid contact through careful consideration of tray design, spacing, and overall height. This ensures the efficient absorption of NOx using sulfuric acid. The inclusion of a turbine (TB-501) to recover energy from the waste gases contributes to reduced operational costs, further improving the feasibility of this plant section

Additionally, the integration of process streams between ammonium nitrate production and waste treatment ensures that the overall system operates efficiently, with sulfuric acid being recycled, reducing overall consumption.

I-core 2.4.2 Key risks and uncertainties

I-core 2.4.2.1 Uncertainties

Several uncertainties were identified during the design process that could affect the overall performance of the system:

T-501(absorption tower): While the design assumptions and Aspen simulations provide a reasonable estimate, the actual efficiency of the gas-liquid contact may differ under real-world conditions. Further testing would help verify the design and ensure optimal NOx removal.

Design: The design is modeled based on lab-scale experiments and Aspen data, which may not perfectly translate to full-scale operations. Using this process may introduce unknown factors that could affect the efficiency of the absorption tower or other equipment. For example, when transitioning to larger equipment sizes, differences in fluid dynamics, mass transfer rates, or heat transfer performance may arise. Additionally, due to variability in upstream processes, changes in the composition of waste gas entering the absorption tower could impact the removal efficiency.

I-core 2.4.2.1 Risks

The main risk in this process is the handling of sulfuric acid, as it poses a significant corrosion risk, particularly in the absorption tower and related equipment (V-501, T-502). Although stainless steel (316) was selected to mitigate this risk, long-term exposure to acidic conditions still threatens the equipment's lifespan. Furthermore, there are inherent safety risks, such as potential leaks, which could lead to major safety incidents.

I-core 2.4.3 Reflection and recommendations for improvement

I-core 2.4.3.1 Reflection

The current design assumes that five stages in the absorption tower will be sufficient for effective NOx removal. However, this is based on simplified design assumptions. A more detailed study of tray spacing and the number of trays could yield better performance and further reduce sulfuric acid consumption. Additionally, the wastewater treatment was not adequately addressed in the design, resulting in the discharge of a significant amount of sulfuric acid, which is certainly not allowed under Queensland's environmental protection regulations. Due to the limitations of the initial design, this issue was not considered, and the environmental burden was calculated in TPO2.3.

I-core 2.4.3.1 Recommendations

A pilot-scale test of the absorption tower would help validate the design assumptions for tray efficiency and overall NOx and N2O removal. This would reduce uncertainty and improve confidence in the plant's long-term performance. Additionally, appropriately increasing the height of the absorption tower and the number of trays would provide sufficient capacity for absorption, ensuring better performance and operational flexibility.

It is recommended to implement a sulfuric acid recovery system in the next phase of the project. This could involve the use of a sulfuric acid recovery tank to capture and recycle sulfuric acid from the process. Alternatively, the wastewater stream could undergo neutralization treatment before discharge, ensuring compliance with environmental regulations and reducing the environmental burden associated with sulfuric acid emissions. The neutralization process can be achieved by using alkaline substances such as sodium hydroxide (NaOH) or calcium hydroxide ($Ca(OH)_2$). When neutralizing sulfuric acid with sodium hydroxide, sodium sulfate (Na_2SO_4) and water are produced, while calcium hydroxide forms calcium sulfate ($CaSO_4$, also known as gypsum) and water. In cases where calcium hydroxide is used, gypsum will precipitate out of the solution and can be removed through settling or filtration. Continuous pH monitoring and control systems should be incorporated to ensure the effluent reaches a neutral pH before discharge. This method of neutralizing acidic wastewater is well-established and has been shown to be effective in mitigating environmental risks, as supported by Akhtar's work on the treatment of acidic wastewater effluents with lime materials (Akhtar, 2022).

I-core 2.4.4 Describe pased decisions

DP1 phase	wrong decision	Corrigendum
	During the early writing of DP1, I initially believed that the absorption design could be	then chose to proceed with Aspen simulations, but since Section 5 is the final unit, its design had to be

	completed through manual calculations. However, after reviewing a large amount of literature on waste gas absorption, I was still unable to obtain effective information.	completed after all the previous sections were designed, which led to many obstacles during the process.
	The initial design did not include a sulfuric acid recycling stream mixed with wastewater, resulting in higher acid consumption and environmental compliance issues due to acid-laden wastewater discharge.	Incorporating sulfuric acid recycling from the beginning would have reduced waste and operational costs, while improving sustainability and compliance with environmental regulations.
	Confusing the concepts of absorption and adsorption, which led to the failure of dp1 and design ambiguity	Understand the concept of both by reading the literature
DP2	Confusing equipment cost with captial cost	Double counting

I-core 2.4.5 Summarize

The design of the waste gas treatment section for the nitric acid and ammonium nitrate production plant successfully addresses key process objectives, such as the removal of nitrogen oxides (NOx and N_2O) and minimizing sulfuric acid consumption. The use of sulfuric acid as an absorbent, combined with effective heat exchange and pumping systems, ensures the plant meets environmental standards while optimizing operational efficiency.

However, the initial decision not to mix the recycle stream with sulfuric acid led to excessive sulfuric acid consumption and reduced removal efficiency. This design choice resulted in higher operating costs and lower process performance. Moving forward, incorporating a sulfuric acid recycling stream or implementing a neutralization system for wastewater is strongly recommended to enhance sustainability and improve both removal efficiency and cost-effectiveness.

In conclusion, while the current design demonstrates feasibility, key improvements—particularly in sulfuric acid recycling and process optimization—will enhance the plant’s overall efficiency and compliance with environmental standards. Implementing these recommendations will ensure the long-term success of the waste gas treatment section and contribute to the plant's sustainability.

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Appendix 1 (T-501)

Hand calculation:

The calculation method comes from **Kister 2024's Distillation.** In total, there are 5 steps to calculate the diameter.

Given data:

Starting value	Value	Units	Description
dH	0.50	inches	Hole diameter
S	35	inches	Tray spacing, inches
Hct	1	inches	Clear Liquid height at the transition from the froth to spray regime, in of liquid.
Cfs	598.9	ft^3/sec	Vapour flow rate (from aspen)
e	71.3		Surface tension (from aspen)
density L =	63.76	lb/ft^3	liquid density
Density gas	0.057	lb/ft^3	Gas density
SF	0.85		Derating factor (assumption SF = 0.85)
Ff	1		Flood factor (assumption Ff = 1)

Step 1 Calculate C-Factor (CSB) using following Kister and Haas Correlation

$$C_{SB} = 0.144\left[\frac{d_H^2\sigma}{\rho_L}\right]^{0.125}\left[\frac{\rho_G}{\rho_L}\right]^{0.1}\left[\frac{S}{h_{ct}}\right]^{0.5}$$

Where,

σ	=	Surface Tension, dyne/cm
ρ_L	=	Liquid Density, lb/ft^3
ρ_G	=	Vapour Density, lb/ft^3

$$Csb = 0.144 * \left(\frac{dh^2 * \delta}{\rho l}\right)^{0.125} * \left(\frac{\rho G}{\rho l}\right)^{0.1} * \left(\frac{S}{hct}\right)^{0.5}$$

CSb = 0.144*0.8526*0.496*5.916 = 0.3044

Step 2 This is the velocity of upward vapour at which liquid droplets are suspended. Calculate Flood Velocity (uN) using following equation:

$$u_N = C_{SB}\sqrt{\frac{\rho_L - \rho_V}{\rho_V}}$$

$$Un = Csb * \sqrt{\frac{\rho l - \rho V}{\rho V}}$$

Un = 0.3044 *33.227 = 11.97629

Step 3 Net Area Calculation:The net area represents smallest area available for vapour flow in the inter-tray spacing. Calculate Net Area (AN) from the flood velocity using following equation

$$A_N = \frac{CFS}{(SF) * F_F * u_N}$$

Where,

CFS = Vapour Flow Rate, ft^3/sec

SF = Derating Factor (See Appendix 2)

F_F = Flood factor, usually taken as 0.8 (80% flood) or designer can choose based on system requirements or project standards

$$An = \frac{CFS}{SF * Ff * Un} = \frac{598.9}{0.85 * 1 * 11.97629} = 58.83192$$

Step 4 Downcomer Area Calculation

$$A_D = \frac{Q_L}{V_{CL}} \qquad Ad = \frac{Ql}{Vcl} = \frac{0.55690}{0.4} = 1.113805$$

QL= Liquid Flow Rate (ft^3/sec) = 0.55690 (from aspen)

Vcl = 0.5, because have spacing = 24 inches = 0.6096 meter = aspen design, and low P

Table: Recommended V_{CL} values for different foaming tendencies

Foaming		V_{CL} in downcomer, ft/s		
		18-in Spacing	24-in Spacing	30-in Spacing
	stabilizers, air-water simulators	0.4-0.5	0.5-0.6	0.5-0.6
Medium	Oil systems, crude oil distillation, absorbers, med. pressure (100-300 psia) hydrocarbon	0.3-0.4	0.4-0.5	0.4-0.5
High	Amine, glycerine, glycols, high-pressure (>300-psi) light hydrocarbons	0.2-0.25	0.2-0.25	0.2-0.3

Step 5 :Tower Diameter Calculation

Tower Diameter = $D_T = \sqrt{\frac{4 * A_T}{\Pi}}$ ıht

TotalTowerArea (AT) = AD + AN

$$At = AD + AN = 58.83192 + 1.113805 = 59.94572$$

$$Dt = \sqrt{4 * \frac{At}{\pi}} = 8.736433\ (ft) = \frac{8.736433}{3.28084} = 2.662865\ (meter)$$

Appendix 2 (E-501 and E-502)

For E-501

Given data:

Q	Aspen data: 483 KW
U	1150 (from table)
Hot in	50.05
Hot out	42
Cold in	30
Cold out	40

$$Q = U * A * delt\ Tlm$$

$$delt\ Tlm = \frac{(50.05 - 40) - (42 - 30)}{\sqrt{\frac{50.05 - 40}{42 - 30}}} = 11$$

To select the U, select water with water. So U = 1150

TABLE 19.1 Typical overall coefficients

Shell and tube exchangers		
Hot fluid	**Cold fluid**	***U*** **(W/m^2°C)**
Heat exchangers		
Water	Water	800–1500
Organic solvents	Organic solvents	100–300
Light oils	Light oils	100–400
Heavy oils	Heavy oils	50–300
Gases	Gases	10–50

$$Area = \frac{Q}{U * delt\ Tlm} = \frac{483 * 1000}{1150 * 11} = 38$$

For E-502

Given data:

Q	Aspen data: 5141 KW
U	1150 (from table)
Hot in	83.08
Hot out	42
Cold in	30
Cold out	40

$$Q = U * A * delt\ Tlm$$

$$delt\ Tlm = \frac{(83.08 - 40) - (42 - 30)}{\sqrt{\frac{83.08 - 40}{42 - 30}}} = 24.32$$

To select the U, select water with water. So U = 1150

TABLE 19.1 Typical overall coefficients

Shell and tube exchangers		
Hot fluid	**Cold fluid**	***U*** **(W/m^2°C)**
Heat exchangers		
Water	Water	800–1500
Organic solvents	Organic solvents	100–300
Light oils	Light oils	100–400
Heavy oils	Heavy oils	50–300
Gases	Gases	10–50

$$Area = \frac{Q}{U * delt\ Tlm} = \frac{5141 * 1000}{1150 * 24.32} = 184\ m\text{^}3$$

Appendix 3 (V-501 and T-502)
For T-502

Given data:

Volumetric flow (cum/sec)	0.00026
Required H2SO4 storage (days)	14 (2 weeks)
Tank freeboard (%)	10

$$Volume = Flow * day * \frac{1000}{1 - \frac{freeboard}{100}} = 0.00026 * 14 * \frac{100}{0.9} = 349\ m^3$$

TABLE 13.2—NOMINAL CAPACITY OF WELDED STORAGE TANKS

Diameter, ft	Height, 20 ft [6.1 m] Capacity		Height, 30 ft [9.1 m] Capacity		Height, 40 ft [12.2 m] Capacity		Height 50 ft [15.2 m] Capacity		Height 60 ft [18.3 m] Capacity		Diameter, m
	Mbbl*	m^{3}**	Mbbl*	m^{3}**	Mbbl*	m^{3}**	Mbbl*	m^{3}*	Mbbl*	m^{3}*	
20	1.1	178	1.7	267	2.2	356	2.8	445	3.4	534	6.1
25	1.7	278	2.6	417	3.5	556	4.4	695	5.2	834	7.6
30	2.5	400	3.8	600	5.0	801	6.3	1,001	7.6	1,201	9.1
35	3.4	545	5.1	817	6.9	1,090	8.6	1,362	10.3	1,635	10.7
40	4.5	712	6.7	1,067	9.0	1,423	11.2	1,779	13.4	2,135	12.2
45	5.7	901	8.5	1,351	11.3	1,801	14.2	2,252	17.0	2,702	13.7
50	7.0	1,112	10.5	1,668	14.0	2,224	17.5	2,780	21.0	3,336	15.2
55	8.5	1,345	12.7	2,018	16.9	2,691	21.2	3,364	25.4	4,036	16.8
60	10.1	1,601	15.1	2,402	20.1	3,202	25.2	4,003	30.2	4,804	18.3
65	11.8	1,879	17.7	2,819	23.6	3,758	29.6	4,698	35.5	5,638	19.8
70	13.7	2,179	20.6	3,269	27.4	4,359	34.3	5,449	41.1	6,538	21.3
80	19.7	2,847	26.9	4,270	35.8	5,693	44.8	7,117	53.7	8,540	24.4
90	22.7	3,603	34.0	5,404	45.3	7,206	56.7	9,007	68.0	10,808	27.4
100	28.0	4,448	42.0	6,672	56.0	8,896	69.9	11,120	83.9	13,344	30.5
110	33.9	5,382	50.8	8,073	67.7	10,764	84.6	13,455	101.6	16,146	33.5
120	40.3	6,405	60.4	9,607	80.6	12,810	100.7	16,012	120.9	19,215	36.6
130	47.3	7,517	70.9	11,275	94.6	15,034	118.2	18,792	141.8	22,551	39.6
140	54.8	8,718	82.3	13,077	109.7	17,436	137.1	21,795	164.5	26,153	42.7
150	62.9	10,008	94.4	15,012	125.9	20,015	157.4	25,019	188.8	30,023	45.7
160	71.6	11,387	107.4	17,080	143.2	22,773	179.1	28,466	214.9	34,160	48.8
180	90.6	14,411	136.0	21,617	181.3	28,822	226.6	36,028	271.9	43,233	54.9
200	111.9	17,791	167.9	26,687	223.8	35,583	279.8	44,479	335.7	53,374	61.0
220	135.4	21,528	203.1	32,291	270.8	43,055	338.5	53,819	406.2	64,583	67.1
240	161.1	25,620	241.7	38,430	322.3	51,239	402.9	64,049	483.4	76,859	73.2
260	189.1	30,068	283.7	45,101	378.2	60,135	472.8	75,169	567.4	90,203	79.3
280	219.3	34,871	329.0	52,307	438.7	69,742	548.4	87,178	658.0	104,614	85.4
300	251.8	40,031	377.7	60,046	503.6	80,062	629.5	100,077	755.4	120,092	91.5

*Mbbl = 1,000 API bbl, **m^3 = cubic meters, API bbl = 42 gal, 1 m^3 = 6.29 bbl

V = 349 m^3, so select diameter = 20 ft = 6.1 meter, height = 40 ft = 12.2 meter

Then the volume = 356 m^3

For T-502

Given data:

Volumetric flow (cum/sec)	0.0176
Required H2SO4 storage (days)	1(days)
Tank freeboard (%)	10

$$Volume = Flow * day * \frac{1000}{1 - \frac{freeboard}{100}} = 0.0176 * 1 * \frac{100}{0.9} = 1690\ m^3$$

V = 1690 m^3, so select diameter = 40 ft = 12.2 meter, height = 50 ft = 15.2 meter

Then the Volume = 1779 m^3

Appendix 4 cost(T-501)

Height of column	3.14	meter
Column diameter	2.721	meter
Column		
wall thickness	0.005	
Shell mass (kg)	**19867.76793**	
a	17400	
b	79	
n	0.85	
Ce(cost in 2010)	**373075.3831**	
Cost index in 2010	550.8	
cost index in 2024	800	
Ce(cost in 2024)	**541866.9326**	
Trays	5	
Tray diameter	2.721	
a	130	
b	550	
n	1.8	
Ce(cost in 2010)	3463.297409	
Cost index in 2010	550.8	
cost index in 2024	800	
Ce(cost in 2024)	5030.206839	
The cost of the trays can be converted to type 304 stainless steel by multiplying by the appropriate materials factor from		
Table 7.6, giving:		
Ce(cost in 2024) with 316 steel	6539.268891	
Final ce(diameter+trays)	548406.2015	



Need to set the wall thickness. For this small column operating at atmospheric pressure, lets use a wall thickness of 5 mm or 0.005 m.

Therefore shell mass:
$(\pi\ (0.22)^2\ -\ \pi\ (0.215)^2)$ x 7 x 7930 = 379 kg.

TABLE 7.6 Materials cost factors, f_m, relative to plain carbon steel

Material	f_m
Carbon steel	1.0
Aluminum and bronze	1.07
Cast steel	1.1
304 stainless steel	1.3
316 stainless steel	1.3
321 stainless steel	1.5
Hastelloy C	1.55
Monel	1.65
Nickel and Inconel	1.7

7.5.2 Cost curves for purchased equipment costs

For those design engineers who lack access to reliable cost data or estimating software, the correlations given in Table 7.2 can be used for preliminary estimates. The correlations in Table 7.2 are of the form:

$$C_e = a + b\,S^n \tag{7.9}$$

where: C_e = purchased equipment cost on a U.S. Gulf Coast basis, Jan. 2010 (CEPCI = 532.9, NF refinery inflation index = 2281.6)
a, b = cost constants in Table 7.2
S = size parameter, units given in Table 7.2
n = exponent for that type of equipment

The correlations in Table 7.2 are only valid between the lower and upper values of S indicated. The prices are all for carbon steel equipment except where noted in the table. Extrapolation to other materials is discussed in Section 7.6.3.

Trays							
Sieve trays	diameter, m	0.5	5.0	130	440	1.8	5
Valve trays	diameter, m	0.5	5.0	210	400	1.9	
Bubble cap trays	diameter, m	0.5	5.0	340	640	1.9	

TABLE 7.2 Purchased equipment cost for common plant equipment—cont'd

Equipment	Units for size, S	S_{lower}	S_{upper}	a	b	n	Note
Pressure vessels							
Vertical, cs	shell mass, kg	160	250,000	11,600	34	0.85	4
Horizontal, cs	shell mass, kg	160	50,000	10,200	31	0.85	4
Vertical, 304 ss	shell mass, kg	120	250,000	17,400	79	0.85	4
Horizontal, 304 ss	shell mass, kg	120	50,000	12,800	73	0.85	4

Appendix 5 cost(T-502 and V-501)

Tank	V-501	T-502
Tank volume	1779	356
a	5800	5800
b	1600	1600
n	0.7	0.7
Ce(cost in 2010)	307269.3	103554.1
Cost index in 2010	550.8	550.8
cost index in 2024	800	800
Ce(cost in 2024)	446288.1	150405.4
Material	PVC	PVC
Factor	0.9	0.9
Ce(cost in 2024) in pvc (AUD)	401659.3	135364.8

Tanks						
Floating roof	capacity, m^3	100	10,000	113,000	3250	0.65
Cone roof	capacity, m^3	10	4000	5800	1600	0.7

7.5.2 Cost curves for purchased equipment costs

For those design engineers who lack access to reliable cost data or estimating software, the correlations given in Table 7.2 can be used for preliminary estimates. The correlations in Table 7.2 are of the form:

$$C_e = a + b\,S^n \tag{7.9}$$

where: C_e = purchased equipment cost on a U.S. Gulf Coast basis, Jan. 2010 (CEPCI = 532.9, NF refinery inflation index = 2281.6)
a, b = cost constants in Table 7.2
S = size parameter, units given in Table 7.2
n = exponent for that type of equipment

The correlations in Table 7.2 are only valid between the lower and upper values of S indicated. The prices are all for carbon steel equipment except where noted in the table. Extrapolation to other materials is discussed in Section 7.6.3.

Appendix 6 Cost (P-501 and P-502)

Compressors type	pump - 501	tb-501	pump - 502
Type	Centrifugal	turnbine	Centrifugal
a	8000	14000	8000
b	240	1900	240
n	0.9	0.75	0.9
duty	2.46	3467.56	0.89
Ce(cost in 2010)	8539.576266	872561.86	8216.103722
Cost index in 2010	550.8	550.8	550.8
cost index in 2024	800	800	800
Ce(cost in 2024) in USA	12403.16088	1267337.5	11933.33874
		Buy	Buy

Pumps and drivers

Single-stage centrifugal	flow liters/s	0.2	126	8000	240	0.9
Explosion proof motor	power, kW	1.0	2500	−1100	2100	0.6
Condensing steam turbine	power, kW	100	20,000	−14,000	1900	0.75

Appendix 7 Cost (E-501 and E-502)

	E-501	E-502
Transfer area	184	38
a	28000	28000
b	54	54
n	1.2	1.2
Ce(cost in 2010)	56195.2	32247.51
Cost index in 2010	550.8	550.8
cost index in 2024	800	800
Ce(cost in 2024) in USA	81619.75	46837.34

Equipment	Units for size, S	S_{lower}	S_{upper}	a	b	n	Note
Exchangers							
U-tube shell and tube	area, m^2	10	1000	28,000	54	1.2	
Floating head shell and tube	area, m^2	10	1000	32,000	70	1.2	
Double pipe	area, m^2	1.0	80	1900	2500	1.0	
Thermosiphon reboiler	area, m^2	10	500	30,400	122	1.1	
U-tube Kettle reboiler	area, m^2	10	500	29,000	400	0.9	
Plate and frame	area, m^2	1.0	500	1600	210	0.95	2